

SMART INDIA HACKATHON 2026

SARA



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2026**

Synchronized Agentic Reasoning & Assistance

- **Problem Statement ID – SIH26047**
- **Problem Statement Title- Patient Case-Taking Software**
- **Theme-MedTech / BioTech / HealthTech**
- **PS Category- Software**
- **Team ID-**
- **Team Name- SynapSix**



IDEA TITLE

Proposed Solution

- SARA is a Biomedical Multi-Agent Decision Support System that analyzes clinical/mystery cases through four specialized AI reasoning agents—AURA, NEXA, LYRA and ITHRA—followed by multi-agent debate, consensus evaluation and final synthesis.

Addressing the problem

- Reduces dependence on a single reasoning perspective
- Enables independent multi-agent analysis
 - Identifies agreements and disagreements through cross-critique
- Produces a structured synthesis with alternative hypotheses and unresolved questions
- Supports users in examining complex clinical cases more systematically



Solution Overview:

4 Specialized Agents
Independent clinical reasoning from different perspectives.

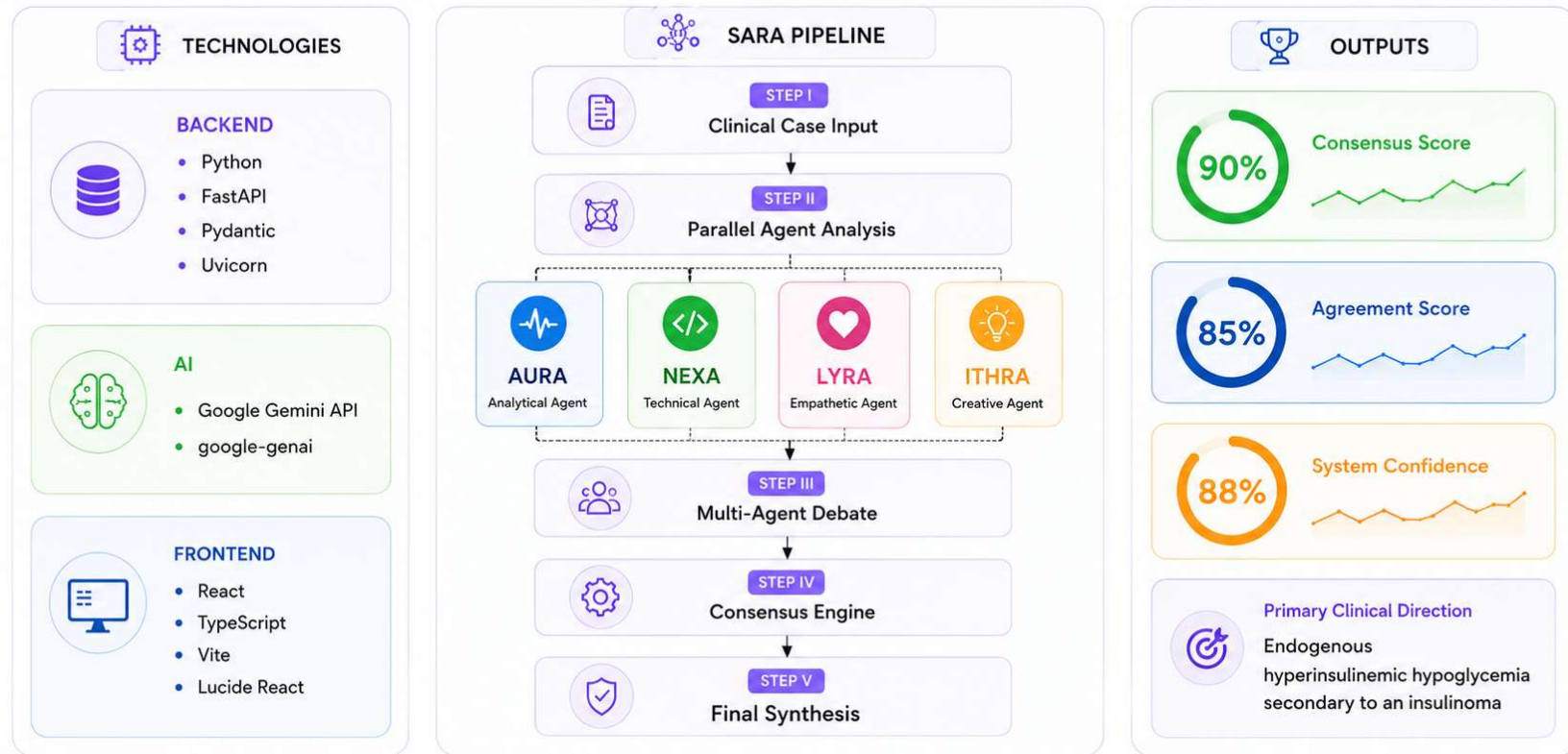
Multi-Agent Debate
Agents critique and refine one another's findings.

Consensus Engine
Aggregates agreement, consensus and system confidence..

Final Synthesis
Generates structured clinical reasoning and recommended investigations.

TECHNICAL APPROACH

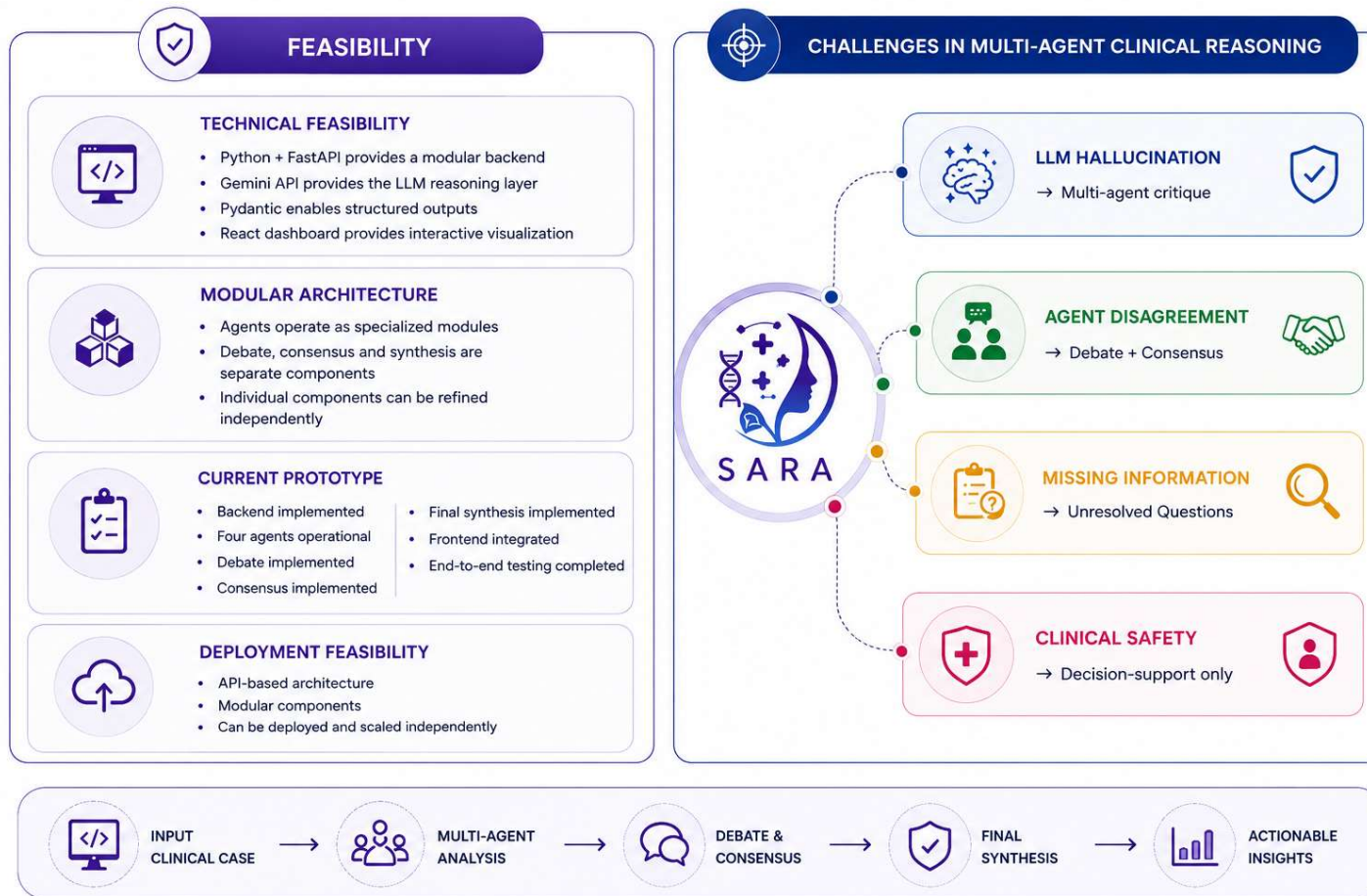
TECHNICAL APPROACH



FEASIBILITY AND VIABILITY



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IMPACT AND BENEFITS



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Impacts

- ✓ Enables structured reasoning for complex clinical cases
- ✓ Provides multiple analytical perspectives
- ✓ Makes agent agreement/disagreement visible
- ✓ Supports transparent examination of alternative hypotheses
- ✓ Helps users identify missing information and possible investigations



Benefits



Clinical / Educational

- Supports structured case reasoning
- Useful for clinical learning and case analysis



Research

- Framework for studying multi-agent biomedical reasoning
- Enables analysis of agent disagreement and consensus



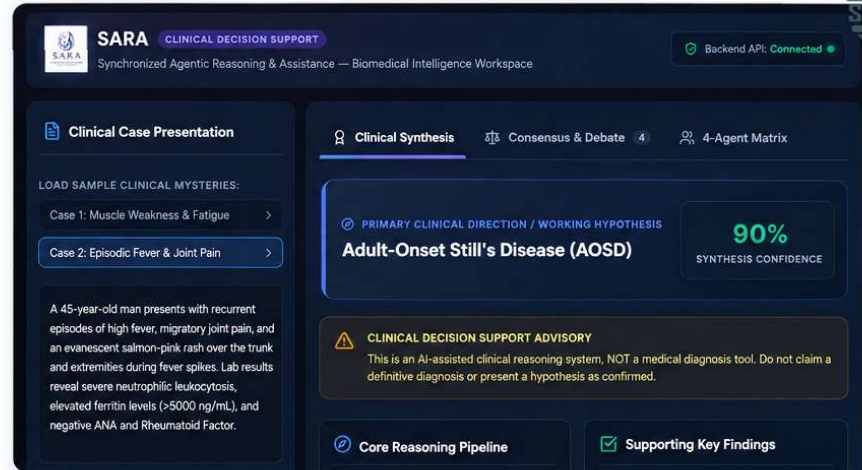
Technical

- Modular multi-agent architecture
- Extensible reasoning pipeline



Scalability

- Can be extended to additional specialized agents
- Potential for multimodal biomedical inputs in future



DEMONSTRATED CASE

Endogenous hyperinsulinemic hypoglycemia secondary to an insulinoma



RESEARCH AND REFERENCES



A. RESEARCH



Research on **Large Language Models** in clinical reasoning



Research on **AI-assisted** clinical decision support



Research on **multi-agent collaborative reasoning**



Research on **debate / critique-based reasoning**



Research on **explainability, uncertainty and trustworthy** healthcare AI



KEY PAPER REFERENCES

1



Batista, A. S. (2025).

Transforming Clinical Practice: The Role of AI-Powered Medical Assistants in Enhancing Healthcare Efficiency and Decision-Making.

Examines AI-powered medical assistants for clinical decision support, workflow efficiency and patient care, while highlighting challenges such as bias, data governance and safe deployment.

[Read Paper](#)

2



Atianashie, M. A., Adaobi, C. C., & Yiadom, C. B. (2026).

Large Language Model-Enabled Health Information Systems: A Systematic Review of FHIR-Based Interoperability, Clinical Decision Support, Privacy, and Trustworthy AI Governance.

Reviews LLM-enabled healthcare systems and highlights clinical decision support, interoperability, privacy, hallucination, governance and the need for human oversight.

[Read Paper](#)

3



Pramila, S., et al. (2026).

Literature Survey on AI Based Healthcare Assistants for Medical Analysis and Personalized Monitoring.

Reviews multimodal AI healthcare assistants for medical analysis and monitoring, while identifying challenges in data scarcity, interpretability and explainability.

[Read Paper](#)

4



Gunawan, A., & Ongowarsito, H. (2026).

Generative Artificial Intelligence in Clinical Practice: Impacts on Medical Decision-Making and Outcomes.

Examines GenAI in clinical decision-making and highlights concerns including bias, privacy, reliability, over-reliance and the need for safe human oversight.

[Read Paper](#)



B. REFERENCES



Google Gemini / Google GenAI Documentation

Official documentation used for Gemini API and the google-genai Python SDK.

<https://ai.google.dev/docs>



FastAPI Documentation

Official documentation used for building SARA's Python REST API and /analyze endpoint.

<https://fastapi.tiangolo.com/>



Pydantic Documentation

Official documentation used for structured schemas, validation and typed API responses.

<https://docs.pydantic.dev/>



React / Vite Documentation

Official documentation used for SARA's interactive frontend and development/build workflow.

<https://react.dev/> | <https://vite.dev/>



Relevant Biomedical / Clinical AI Literature

Research literature covering clinical decision support, healthcare AI, LLMs, explainability and trustworthy AI.

<https://pubmed.ncbi.nlm.nih.gov/>



SIH Problem Statement / Domain References

Official Smart India Hackathon problem statement and relevant biomedical / healthcare domain references.

<https://sih.gov.in/>

FUTURE SCOPE

